THE GENERALIZED STOKES THEOREM AND BOUNDARY DATA: NEW NEW METHODS FOR AREA AND VOLUME MEASUREMENTS ON CURVED SURFACES

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Abstract. We derive a set of combinatorial formulas for computing the area or volume enclosed by a collection of ordered data points on surfaces with non-euclidean geometries. The majority of the formulas presented in this work can be applied analytically, and, to the authors knowledge, have not yet been presented elsewhere in the literature. The driving force behind our derivations is the generalized Stokes theorem expressed in the language of differential forms. In this preliminary work, the mathematical formalism for measuring area and volume enclosed by linear piece-wise loops on curved surfaces is developed and demonstrated for a selection of common geometrical objects and coordinate systems. Future work will consider specific applications of the derived formulas, including comparisons with more computationally intensives techniques.

1. Introduction

Medical technicians, geologist, astronomers, and data scientists are all concerned with obtaining accurate area and volume measurements from data. The "surveyors area formula" [2] is a classical, discrete version of Green's theorem for computing the area enclosed by the piecewise-linear boundary of a two-dimensional surface in a Cartesian coordinate system. This technique is easily extended with the divergence theorem, resulting in a combinatorial formula for the volume enclosed by a surface covered with an oriented triangular mesh [4]. In scenarios where area and volume estimates are necessary, these formulas are considered standard mathematics, with incremental upgrades yielding more accurate—though sometimes computationally expensive—methods for approximating the area and volume enclosed by various surfaces (see for example [5, 6, 7]).

Most techniques for estimating area and volume from data do not account for non-euclidean curvature. Some works, however, do make efforts to include the effects of surface curvature on embedded surface area. In particular, [3] uses results from spherical geometry in order to derive a number of exact and approximate formulas for measurements involving polygons on spheres. This work is extended by [1], where an exact formula for the area enclosed by a linear-piecewise loop on the sphere is derived naturally with differential forms.

In this work, we extend the ideas of [1] and derive a set of exact combinatorial formulas, most of which can be employed analytically, for measuring the area and volume enclosed by boundary data on common surfaces with non-euclidean geometries. The driving force behind our work is the generalized Stokes theorem expressed in the language of differential forms. While not commonly used in the

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context of applied mathematics, differential forms provide a natural basis for the subject of this paper compared to the classical vector calculus techniques (e.g. *div*, *grad*, *curl*).

The paper is organized as follows. In §2 we establish the generalized Stokes theorem, in §3 we develop the relevant formalism, and in §4 we apply the formalism to numerous surfaces and coordinate systems.

2. The Generalized Stokes Theorem

We introduce the basis k-form on $\mathbb{R}^n$ as $dx_1 \wedge dx_2 \wedge ... \wedge dx_k$. Let ω be a k-form so that $d\omega$ is a $(k+1)$ form. Gauss', Green's, and the 2-dimensional Stokes theorem are all captured by the generalized Stokes theorem:

$$\int_{\partial D} \omega = \int_D d\omega \tag{1}$$

for D, a smooth $(k+1)$ dimensional manifold on $\mathbb{R}^n$ with k dimensional boundary, ∂D.

3. Formalism

3.1. **2-Dimensional Surface, 1-Dimensional Boundary.** Let S be a simply connected 2-dimensional surface embedded in $\mathbb{R}^3$ with boundary ∂S. Suppose S is naturally parametrized by the coordinates (u, v), such that

$$\mathbf{G}(u,v) = \begin{pmatrix} x(u,v) \\ y(u,v) \\ z(u,v) \end{pmatrix}, \tag{2}$$

where $\mathbf{G}$ denotes the position vector extending from the origin to a point on S. Suppose that ∂S is composed of N piecewise-linear segments, individually denoted by $\boldsymbol{\gamma_i}$. Collectively, the $\boldsymbol{\gamma_i}$'s form a closed, oriented loop. Let each segment $\boldsymbol{\gamma_i}$ have end and start points denoted by $\left(u_1^i, v_1^i\right)$ and $\left(u_2^i, v_2^i\right)$, respectively. We parametrize each linear segment $\boldsymbol{\gamma_i}$ in the (u, v) plane with

$$\boldsymbol{\gamma_i}(t) = \begin{pmatrix} u(t) \\ v(t) \end{pmatrix} = \begin{pmatrix} u_1^i + (u_2^i - u_1^i)t \\ v_1^i + (v_2^i - v_1^i)t \end{pmatrix};\ 0 < t < 1. \tag{3}$$

The ith boundary segment of S is then parametrized by $\mathbf{G}\left(\gamma_{\mathbf{i}}(t)\right)$. Differentiating with respect to t, the derivative of $\boldsymbol{\gamma_i}$ is

$$D\boldsymbol{\gamma_i} = \begin{pmatrix} u_2^i - u_1^i \\ v_2^i - v_1^i \end{pmatrix}. \tag{4}$$

Next, we define ω so that $d\omega$ is the area 2-form on S such that

$$Area(S) = \int_S d\omega. \tag{5}$$

The area form on S is given by the magnitude of the cross product

$$d\omega = \left\| \frac{d\boldsymbol{G}}{du} \times \frac{d\boldsymbol{G}}{dv} \right\| du \wedge dv. \tag{6}$$

By the generalized Stokes theorem (Eq. 1),

$$Area(S) = \int_S d\omega = \int_{\partial S} \omega = \sum_{i=1}^{N} \int_{\boldsymbol{\gamma_i}} \omega. \tag{7}$$

Since ω is a 1-form, the integral insde the summation in Eq. 7 reduces to a path integral. Assuming ω is of the form

$$\omega = f(u,v)\,du + g(u,v)\,dv, \tag{8}$$

then the integral of ω over the linear segment $\boldsymbol{\gamma_i}$ is

$$\int_{\boldsymbol{\gamma_i}} \omega = \int_0^1 \begin{pmatrix} f(\boldsymbol{\gamma_i}(t)) \\ g(\boldsymbol{\gamma_i}(t)) \end{pmatrix} \cdot D\boldsymbol{\gamma_i}\ dt, \tag{9}$$

which reduces to

$$\int_{\boldsymbol{\gamma_i}} \omega = \int_0^1 \Big[f\left(\boldsymbol{\gamma_i}(t)\right)(u_2^i - u_1^i) + g\left(\boldsymbol{\gamma_i}(t)\right)(v_2^i - v_1^i) \Big] dt. \tag{10}$$

Finally, using Eq. 7 we obtain a general expression for the area enclosed by a piece-wise loop embedded on the surface S. Namely,

$$Area(S) = \sum_{i=1}^{N} \int_0^1 \Big[f\left(\boldsymbol{\gamma_i}(t)\right)(u_2^i - u_1^i) + g\left(\boldsymbol{\gamma_i}(t)\right)(v_2^i - v_1^i) \Big] dt. \tag{11}$$

3.2. **3-Dimensional Manifold, 2-Dimensional Boundary.** Let D be a simple, smooth, closed, solid region in $\mathbb{R}^3$ naturally parametrized in (u,v,w) coordinates, such that

$$\mathbf{G}(u,v,w) = \begin{pmatrix} x(u,v,w) \\ y(u,v,w) \\ z(u,v,w) \end{pmatrix}, \tag{12}$$

where $\mathbf{G}$ denotes the position vector extending from the origin to a point in D. Suppose the boundary of the solid, ∂D, in (u,v,w) space is covered by a triangular mesh comprised of N triangles, labeled T^i. Suppose each T^i has vertices $\mathbf{A}_j^i \equiv \left(u_j^i, v_j^i, w_j^i\right)$ for $j = 1,2,3$ (3 vertices), cyclically ordered so as to induce a positive orientation on the surface with an outwards pointing normal. Each triangular surface T^i can be linearly parametrized as

$$\boldsymbol{\gamma_i}(t_1,t_2) = \begin{pmatrix} u(t_1,t_2) \\ v(t_1,t_2) \\ w(t_1,t_2) \end{pmatrix} = \mathbf{A}_1^i + \left(\mathbf{A}_2^i - \mathbf{A}_1^i\right) t_1 + \left(\mathbf{A}_3^i - \mathbf{A}_1^i\right) t_2 \tag{13}$$

for $0 < t_1 < 1 - t_2$ and $0 < t_2 < 1$. The Jacobian is

$$D\boldsymbol{\gamma_i} = \left(\mathbf{A}_2^i - \mathbf{A}_1^i, \quad \mathbf{A}_3^i - \mathbf{A}_1^i\right). \tag{14}$$

Next, we define β so that $d\beta$ is the volume 3-form on D such that

$$Vol(D) = \int_D d\beta. \tag{15}$$

The volume 3-form can be derived from the determinant of the Jacobian of $\boldsymbol{G}$, i.e. $d\beta = \left|\det\left(D\boldsymbol{G}\right)\right| du \wedge dv \wedge dw$. By the generalized Stokes theorem (Eq. 1),

$$Vol(D) = \int_D d\beta = \int_{\partial D} \beta = \sum_{i=1}^{N} \int_{\boldsymbol{\gamma_i}} \beta\left(D\boldsymbol{\gamma_i}\right), \tag{16}$$

where the summation runs over each triangular surface, parametrized by $\boldsymbol{\gamma_i}$. Since β is a 2-form, the integral in the summation term in Eq. 16 reduces to a surface integral over each T^i. Assuming β is of the form

$$\beta = f(u,v,w)\ du \wedge dv + g(u,v,w)\ dv \wedge dw + h(u,v,w)\ dw \wedge du, \tag{17}$$

then the integral of β over a single T^i can be written as

$$\begin{aligned}\int_{\boldsymbol{\gamma}_i} \beta\left(D\boldsymbol{\gamma_i}\right) = \int_0^1 \int_0^{1-t_2} \Bigg[& f\left(\boldsymbol{\gamma_i}(t_1,t_2)\right) \begin{vmatrix} (\mathbf{A}_2^i - \mathbf{A}_1^i)_u & (\mathbf{A}_3^i - \mathbf{A}_1^i)_u \\ (\mathbf{A}_2^i - \mathbf{A}_1^i)_v & (\mathbf{A}_3^i - \mathbf{A}_1^i)_v \end{vmatrix} \\ & + g\left(\boldsymbol{\gamma_i}(t_1,t_2)\right) \begin{vmatrix} (\mathbf{A}_2^i - \mathbf{A}_1^i)_v & (\mathbf{A}_3^i - \mathbf{A}_1^i)_v \\ (\mathbf{A}_2^i - \mathbf{A}_1^i)_w & (\mathbf{A}_3^i - \mathbf{A}_1^i)_w \end{vmatrix} \\ & + h\left(\boldsymbol{\gamma_i}(t_1,t_2)\right) \begin{vmatrix} (\mathbf{A}_2^i - \mathbf{A}_1^i)_w & (\mathbf{A}_3^i - \mathbf{A}_1^i)_w \\ (\mathbf{A}_2^i - \mathbf{A}_1^i)_u & (\mathbf{A}_3^i - \mathbf{A}_1^i)_u \end{vmatrix} \Bigg] dt_1 dt_2,\end{aligned} \tag{18}$$

where $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is the determinant $(ad - bc)$, and $(\mathbf{A}_j^i - \mathbf{A}_k^i)_\ell$ denotes the ℓth component of the vector difference $\mathbf{A}_j^i - \mathbf{A}_k^i$. Finally, using Eq. 16 we obtain a general formula for the volume enclosed by a non-euclidean triangular mesh parametrized by $\boldsymbol{\gamma_i}$. Namely,

$$\begin{aligned}Vol(D) = \sum_{i=1}^{N} \int_0^1 \int_0^{1-t_2} \Bigg[& f\left(\boldsymbol{\gamma_i}(t_1,t_2)\right) \begin{vmatrix} (\mathbf{A}_2^i - \mathbf{A}_1^i)_u & (\mathbf{A}_3^i - \mathbf{A}_1^i)_u \\ (\mathbf{A}_2^i - \mathbf{A}_1^i)_v & (\mathbf{A}_3^i - \mathbf{A}_1^i)_v \end{vmatrix} \\ & + g\left(\boldsymbol{\gamma_i}(t_1,t_2)\right) \begin{vmatrix} (\mathbf{A}_2^i - \mathbf{A}_1^i)_v & (\mathbf{A}_3^i - \mathbf{A}_1^i)_v \\ (\mathbf{A}_2^i - \mathbf{A}_1^i)_w & (\mathbf{A}_3^i - \mathbf{A}_1^i)_w \end{vmatrix} \\ & + h\left(\boldsymbol{\gamma_i}(t_1,t_2)\right) \begin{vmatrix} (\mathbf{A}_2^i - \mathbf{A}_1^i)_w & (\mathbf{A}_3^i - \mathbf{A}_1^i)_w \\ (\mathbf{A}_2^i - \mathbf{A}_1^i)_u & (\mathbf{A}_3^i - \mathbf{A}_1^i)_u \end{vmatrix} \Bigg] dt_1 dt_2.\end{aligned} \tag{19}$$

4. Applications

4.1. **Area on the Pseudo-Sphere.** The pseudo-sphere with constant negative-Gaussian curvature has the parametrization

$$\mathbf{G}\left(u,v\right) = \begin{pmatrix} \operatorname{sech} u \cos v \\ \operatorname{sech} u \sin v \\ u - \tanh u, \end{pmatrix}$$

for $u \in (-\infty, \infty)$ and $v \in [0, 2\pi)$. The area 2-form is given by

$$d\omega = -\operatorname{sech} u \tanh u\ du \wedge dv.$$

A suitable choice for ω is $\omega = \operatorname{sech} u \; dv$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = \int_0^1 (v_2^i - v_1^i) \operatorname{sech}\left(u_1^i + (u_2^i - u_1^i)t\right) dt$$

and sum over each linear segment in the (u, v) plane to find the area enclosed by a piece-wise loop (∂S):

$$Area(S) = \sum_i \frac{v_2^i - v_1^i}{u_2^i - u_1^i} \left[\tan^{-1}\left(\sinh u_2^i\right) - \tan^{-1}\left(\sinh u_1^i\right)\right]. \tag{20}$$

An illustration is provided in Fig. 1.

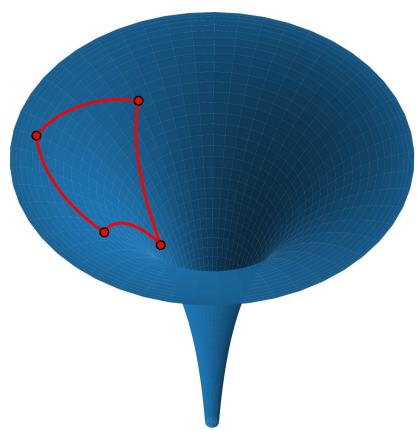

FIGURE 1. The pseudo-sphere and a piece-wise loop. Eq. 20 provides an exact expression for the area enclosed by an arbitrary piece-wise loop on the pseudo-sphere given the end and start points of each piece-wise segment.

4.2. **Area on the Torus.** The azimuthally symmetric torus has the cartesian equation

$$\left(c - \sqrt{x^2 + y^2}\right)^2 + z^2 = a^2, \tag{21}$$

where c is the radius extending from the center of the hole to the center of the torus, and a is the radius of the tube. The torus has the parametrization

$$\mathbf{G}(u, v) = \begin{pmatrix} \left(c + a\cos(v)\right)\cos(u) \\ \left(c + a\cos(v)\right)\sin(u) \\ a\sin(v) \end{pmatrix} \tag{22}$$

for $u, v \in [0, 2\pi)$. The area 2-form is

$$d\omega = a\left(c + a\cos(v)\right) du \wedge dv, \tag{23}$$

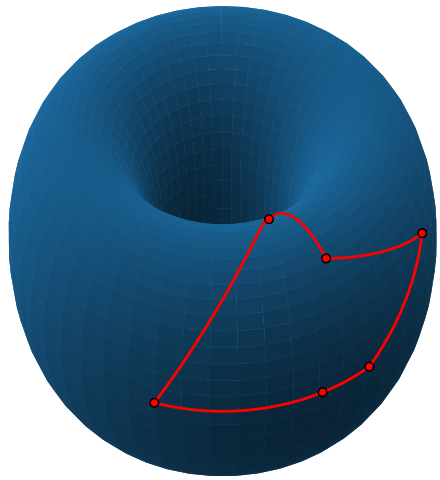

FIGURE 2. The torus and a piece-wise loop. Eq. 25 provides an exact expression for the area enclosed by an arbitrary piece-wise loop on the torus given the end and start points of each piece-wise segment.

where a suitable choice for ω is $\omega = a\left(c + a\cos(v)\right)u\ dv$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = \int_0^1 a\left(c + a\cos\left(v_1^i + (v_2^i - v_1^i)t\right)\right)\left(u_1^i + (u_2^i - u_1^i)t\right)\left(v_2^i - v_1^i\right) dt \tag{24}$$

and sum over each linear segment in the (u, v) plane to find the area enclosed by a piece-wise loop (∂S):

$$\begin{aligned} Area(S) = \sum_i \frac{a}{2\left(v_2^i - v_1^i\right)} \Bigg[& 2a\left(u_1^i - u_2^i\right)\left(\cos\left(v_1^i\right) - \cos\left(v_2^i\right)\right) \\ & + \left(v_1^i - v_2^i\right)\left(c\left(u_1^i + u_2^i\right)\left(v_1^i - v_2^i\right) + 2a\left(u_1^i \sin\left(v_1^i\right) - u_2^i \sin\left(v_2^i\right)\right)\right)\Bigg]. \end{aligned} \tag{25}$$

An illustration is provided in Fig. 2

4.3. **Area on the Paraboloid.** The paraboloid $z = x^2 + y^2$ is naturally parametrized in cylindrical coordinates:

$$\mathbf{G}(r, \theta) = \begin{pmatrix} r\cos\theta \\ r\sin\theta \\ r^2 \end{pmatrix}$$

for $r > 0$ and $\theta \in [0, 2\pi)$. The area 2-form is

$$d\omega = \left(4r^4 + r^2\right)\ d\theta \wedge dr,$$

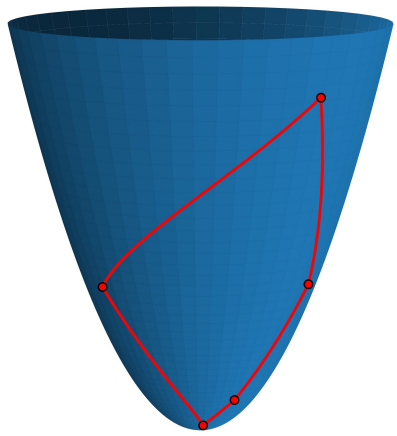

FIGURE 3. The paraboloid and a piece-wise loop. Eq. 26 provides an exact expression for the area enclosed by an arbitrary piece-wise loop on the paraboloid given the end and start points of each piece-wise segment.

where a suitable choice for ω is $\omega = r^2\theta\left(4r^2+1\right)dr$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = \int_0^1 \Big[\left(\theta_1^i + \Delta\theta^i t\right)\left(4\left(r_1^i + \Delta r^i t\right)^4 + \left(r_1^i + \Delta r^i t\right)^2\right)\Delta r^i \Big] dt$$

where $\Delta r^i \equiv r_2^i - r_1^i$ and $\Delta\theta^i \equiv \theta_2^i - \theta_1^i$. We sum over each linear segment in the (r,θ) plane to find the area enclosed by a piece-wise loop (∂S):

$$\begin{aligned} Area(S) = \sum_i \Big[\frac{\Delta r^i}{60} \Big(& 30 r_1^{i^2}\left(1+4r_1^{i^2}\right)\left(\Delta\theta^i + 2\theta_1^i\right) + 20\Delta r^i\left(r_1^i + 8 r_1^{i^3}\right) \\ & \times \left(2\Delta\theta^i + 3\Delta\theta_1^i\right) + 5\left(\Delta r^i\right)^2\left(1+24r_1^{i^2}\right)\left(3\Delta\theta^i + 4\theta_1^i\right) \\ & + 48\left(\Delta r^i\right)^3 r_1^i\left(4\Delta\theta^i + 5\theta_1^i\right) + 8\left(\Delta r^i\right)^4\left(5\Delta\theta^i + 6\theta_1^i\right)\Big)\Big]. \end{aligned} \tag{26}$$

An illustration is provided in Fig. 3.

4.4. **Area on the Hyperboloid.** The one-sheeted circular hyperboloid, oriented along the z-axis has the cartesian equation $\frac{x^2+y^2}{a^2} - \frac{z^c}{c^2} = 1$. We parametrize this surface with

$$\boldsymbol{G}(u,v) = \begin{pmatrix} a\sqrt{1+u^2}\cos v \\ a\sqrt{1+u^2}\sin v \\ cu \end{pmatrix}$$

for $u \in \mathbb{R}$ and $v \in [0, 2\pi)$. The area 2-form is

$$d\omega = a\sqrt{c^2\left(1+u^2\right) + a^2u^2}\; du \wedge dv,$$

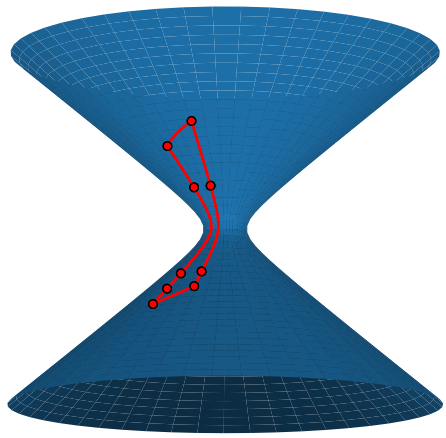

FIGURE 4. The one-shetted circular hyperboloid and a piece-wise loop. Eq. 27 provides an exact expression for the area enclosed by an arbitrary piece-wise loop on the hyperboloid given the end and start points of each piece-wise segment.

where a suitable choice for $\omega = av\sqrt{c^2\left(1+u^2\right)+a^2u^2}\ du$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = a\int_0^1 \left(v_1^i + \left(v_2^i - v_1^i\right)t\right)\sqrt{c^2\left(1+\left(u_1^i+\left(u_2^i-u_1^i\right)t\right)^2\right)+a^2\left(u_1^i+\left(u_2^i-u_1^i\right)t\right)^2}\,dt$$

and sum over each linear segment in the (u,v) plane to find the area enclosed by a piece-wise loop (∂S):

$$\begin{aligned}
&(27)\quad Area(S) = \\
&\sum_i \frac{a}{2\left(\Delta u^i\right)^2}\Bigg\{\Bigg[\frac{-1}{3}\sqrt{a^2u_1^{i^2}+c^2\left(1+u_1^{i^2}\right)}\left(\frac{2c^2\Delta v^i}{a^2+c^2}+u_1^i\left(-\Delta v^iu_1^i\left(-\Delta v^iu_1^i+3\Delta u^i+v_1^i\right)\right)\right)\Bigg] \\
&+\frac{1}{3}\sqrt{a^2\left(\Delta u^i+u_1^i\right)^2+c^2\left(1+\left(\Delta u^i+u_1^i\right)^2\right)}\left[\frac{2c^2\Delta v^i}{a^2+c^2}+\left(\Delta u^i+u_1^i\right)\left(2\Delta u^i\Delta v^i-\Delta v^iu_1^i+3\Delta u^iv_1^i\right)\right] \\
&+\frac{c^2\left(\Delta v^iu_1^i-\Delta u^iv_1^i\right)}{\sqrt{a^2+c^2}}\left(\log\left[\left(a^2+c^2\right)u_1^i+\sqrt{\left(a^2+c^2\right)\left(a^2u_1^{i^2}+c^2\left(1+u_1^{i^2}\right)\right)}\right]-1\right) \\
&+\log\left[\left(a^2+c^2\right)\left(\Delta u^i+u_1^i\right)+\sqrt{\left(a^2+c^2\right)\left(a^2\left(\Delta u^i+u_1^i\right)^2+c^2\left(1+\left(\Delta u^i+u_1^i\right)^2\right)\right)}\right]\Bigg\},
\end{aligned}$$

where $\Delta u^i \equiv u_2^i - u_1^i$ and $\Delta v^i \equiv v_2^i - v_1^i$. An illustration is provided in Fig. 4.

4.5. **Area on the Parabolic Cylinder.** The parabolic cylinder $z = x^2$ has the parametrization

$$\boldsymbol{G}\left(u,v\right) = \begin{pmatrix} u \\ v \\ u^2 \end{pmatrix}$$

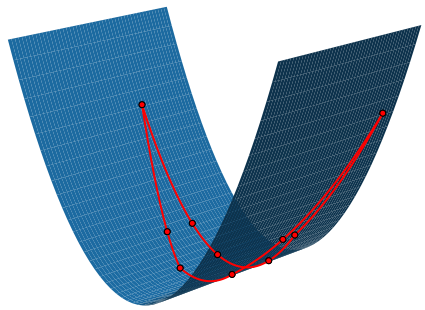

FIGURE 5. The parabolic cylinder and a piece-wise loop. Eq. 28 provides an exact expression for the area enclosed by an arbitrary piece-wise loop on the parabolic cylinder given the end and start points of each piece-wise segment.

for $u, v \in \mathbb{R}$. The area 2-form is given by

$$d\omega = \sqrt{1+4u^2}\ du \wedge dv,$$

where a suitable choice for ω is $\omega = -v\sqrt{1+4u^2}\ du$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = -\int_0^1 \left(v_1^i + \Delta v^i t\right) \sqrt{1+4\left(u_1^i + \Delta u^i t\right)^2} \Delta u^i\ dt$$

where $\Delta u^i \equiv u_2^i - u_1^i$ and $\Delta v^i \equiv v_2^i - v_1^i$. We sum over each linear segment in the (u, v) plane to find the area enclosed by a piece-wise loop (∂S):

$$\begin{aligned} Area(S) = \sum_i \frac{1}{12\Delta u^i} \Bigg[& \left(\Delta v^i - 2\Delta v^i {u_1^i}^2 + 6\Delta u^i u_1^i v_1^i\right) \\ & \sqrt{1+4{u_1^i}^2} + 3\left(\Delta u^i v_1^i - \Delta v^i u_1^i\right) \operatorname{arcsinh}\left(2u_1^i\right) \\ - \Bigg(& \sqrt{1+4\left(\Delta u^i + u_1^i\right)^2} \left(\Delta v^i + 2\Delta v^i \left(2\Delta u^i - u_1^i\right)\left(\Delta u^i + u_1^i\right) + 6\Delta u^i \left(\Delta u^i + u_1^i\right) v_1^i\right) \\ & + 3\left(\Delta u^i v_1^i - \Delta v^i u_1^i\right)\Bigg) \operatorname{arcsinh}\left(2\left(\Delta u^i + u_1^i\right)\right)\Bigg]. \end{aligned} \tag{28}$$

An illustration is provided in Fig. 5.

4.6. **Area on the Ellipsoid.** The ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ has the parametrization

$$\boldsymbol{G}(u, v) = \begin{pmatrix} a\cos u \sin v \\ b \sin u sinv \\ c \cos v \end{pmatrix}$$

for $u \in [0, 2\pi)$, $v \in [0, \pi]$, and $a, b, c > 0$. The area 2-form is given by

$$d\omega = \sqrt{a^2c^2\sin^4 v \sin^2 u + b^2c^2 \sin^4 v \cos^2 u + a^2b^2 \sin^2 v \cos^2 v}\ du \wedge dv.$$

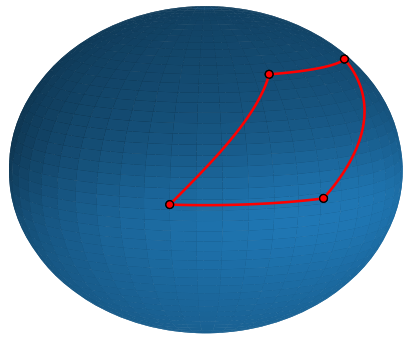

FIGURE 6. The ellipsoid $\frac{x^2+y^2}{a^2} + \frac{z^2}{c^2} = 1$ and a piece-wise loop. Eq. 29 provides an exact expression for the area enclosed by an arbitrary piece-wise loop on the ellipsoid given the end and start points of each piece-wise segment. The expression can be evaluated numerically.

Assuming the special case for which $a = b \equiv Q$, the area 2-form reduces to

$$d\omega = Q \sin v \sqrt{c^2 \sin^2 v + Q^2 \cos^2 v}\ du \wedge dv.$$

A suitable choice for ω is $\omega = Qu \sin v \sqrt{c^2 \sin^2 v + Q^2 \cos^2 v}\ dv$. Integrating this 1-form over a closed loop is not analytically tractable as in the previous cases, but is numerically feasible. Parametrizing the ith line segment joining the point (u_1^i, v_1^i) to (u_2^i, v_2^i), we find

$$Area(S) = \sum_i \int_0^1 \Delta v^i Q \left(u_1 + \Delta u^i t\right) \sin\left(v_1^i + \Delta v^i t\right) \sqrt{c^2 \sin^2\left(v_1^i + \Delta v^i t\right) + Q^2 \cos^2\left(v_1^i + \Delta v^i t\right)}\ dt. \tag{29}$$

4.7. **Area on the Cylinder.** The cylinder of radius R has the cylindrical coordinate parametrization

$$\boldsymbol{G}(\theta, z) = \begin{pmatrix} R\cos\theta \\ R\sin\theta \\ z \end{pmatrix}$$

for $\theta \in [0, 2\pi)$ and $z \in \mathbb{R}$. The area 2-form is given by

$$d\omega = R^2\ d\theta \wedge dz$$

where a suitable choice for ω is $\omega = -R^2 z\ d\theta$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = \int_0^1 -R^2 \left(z_1^i + \left(z_2^i - z_1^i\right) t\right) \left(\theta_2^i - \theta_1^i\right) dt$$

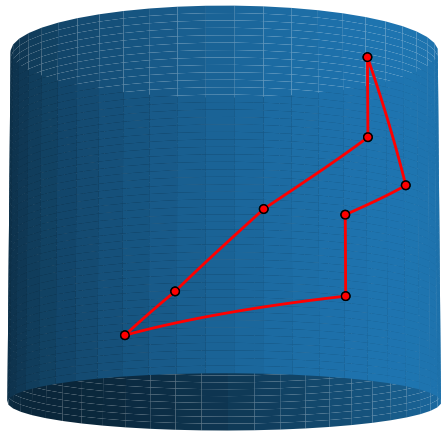

FIGURE 7. The cylinder and a piece-wise loop. Eq. 30 provides an exact expression for the area enclosed by an arbitrary piece-wise loop on the cylinder given the end and start points of each piece-wise segment.

and sum over each linear segment in the (θ, z) plane to find the area enclosed by a piece-wise loop (∂S):

$$Area(S) = \sum_i \frac{\theta_1^i - \theta_2^i}{2} R^2 \left(z_1^i + z_2^i \right). \tag{30}$$

An illustration is provided in Fig. 7.

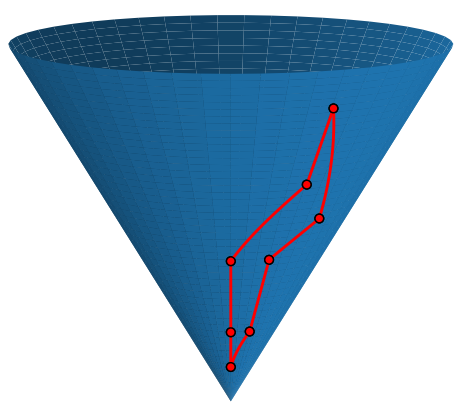

FIGURE 8. The cone and a piece-wise loop. Eq. 32 provides an exact expression for the area enclosed by an arbitrary piece-wise loop on the cone given the end and start points of each piece-wise segment.

4.8. **Area on the Cone.** The upper half of the cone, $z = \sqrt{x^2+y^2}$, can be parametrized in cylindrical coordinates, for which

$$\mathbf{G}(r,\theta) = \begin{pmatrix} r\cos\theta \\ r\sin\theta \\ r \end{pmatrix}$$

for $r > 0$ and $\theta \in [0, 2\pi)$. In the (r,θ) plane, the area 2-form is given by

$$d\omega = r\sqrt{2}\ dr \wedge d\theta$$

where a suitable choice for ω is $\omega = \frac{\sqrt{2}}{2} r^2 d\theta$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = \int_0^1 \frac{\sqrt{2}}{2}\left(r_1^i + \left(r_2^i - r_1^i\right)t\right)^2 \left(\theta_2^i - \theta_1^i\right) dt \tag{31}$$

and sum over each linear segment in the (r,θ) plane to find the area enclosed by a piece-wise loop (∂S):

$$Area\,(S) = \frac{\sqrt{2}}{6}\sum_i \left(\theta_2^i - \theta_1^i\right)\left(r_1^{i^2} + r_1^i r_2^i + r_2^{i^2}\right). \tag{32}$$

An illustration is provided in Fig. 8. Alternatively, for the more general case $z = c\sqrt{x^2+y^2}$, a spherical coordinate parametrization of this surface gives us yet another formula for $Area(S)$. In particular, we parametrize the cone with

$$\mathbf{G}(\rho,\theta) = \begin{pmatrix} a\rho\cos\theta \\ a\rho\sin\theta \\ b\rho \end{pmatrix}$$

where $a = \sin\phi_0,\ b = \cos\phi_0$ and $\phi_0 = \arctan\left(1/\mathrm{c}\right)$ for $c > 0$. We also have that $\rho \geq 0$, and $\theta \in [0, 2\pi)$. In this coordinate system, the area 2-form is

$$d\omega = a\rho\sqrt{a^2+b^2}\ d\rho \wedge d\theta,$$

where a suitable choice for ω is $\omega = \frac{1}{2}\rho^2 a\sqrt{a^2+b^2}\ d\theta$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = \frac{a}{2}\sqrt{a^2+b^2}\int_0^1 \left(\rho_1^i + \left(\rho_2^i - \rho_1^i\right)t\right)^2 \left(\theta_2^i - \theta_1^i\right)\ dt.$$

and sum over each linear segment in the (ρ,θ) plane to find the area enclosed by a piece-wise loop (∂S):

$$Area(S) = \frac{a\sqrt{a^2+b^2}}{6}\sum_i \left(\theta_2^i - \theta_1^i\right)\left(\rho_1^{i^2} + \rho_1^i\rho_2^i + \rho_2^{i^2}\right). \tag{33}$$

4.9. **Polar Coordinates.** The polar coordinate system is defined by the mapping

$$\mathbf{G}(r,\theta) = \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix}$$

for $r > 0$ and $\theta \in [0, 2\pi)$. The area 2-form in this coordinate system is

$$d\omega = r\ dr \wedge d\theta,$$

where a suitable choice for ω is

$$\omega = \frac{1}{2} r^2 d\theta. \tag{34}$$

Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = \int_0^1 \frac{\theta_2^i - \theta_1^i}{2} \left(r_1^i + (r_2^i - r_1^i)t \right)^2 dt$$

and sum over each linear segment in the (r, θ) plane to find the area enclosed by a piece-wise loop (∂S):

$$Area\,(S) = \sum_i \frac{\theta_2^i - \theta_1^i}{6} \left(r_1^{i^2} + r_1^i r_2^i + r_2^{i^2} \right). \tag{35}$$

4.10. **Elliptical Coordinates.** The elliptical coordinate system is defined by the mapping

$$\mathbf{G}\,(u, v) = \begin{pmatrix} a \cosh u \cos v \\ a \sinh u \sin v \end{pmatrix}$$

for $u \geq 0$ and $v \in [0, 2\pi)$. In this coordinate system, the area 2-form is

$$d\omega = \frac{a^2}{2} \left(\cosh(2u) - \cos(2v) \right) \; du \wedge dv$$

where a suitable choice for ω is $\omega = \frac{a^2}{4} \left(\sinh(2u) \; dv + \sin(2v) \; du \right)$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = \frac{a^2}{4} \int_0^1 \left(\sinh\left[2 \left(u_1^i + (u_2^i - u_1^i)t \right) \right] \left(v_2^i - v_1^i \right) + \sin\left[2 \left(v_1^i + (v_2^i - v_1^i)t \right) \right] \left(u_2^i - u_1^i \right) \right) dt$$

and sum over each linear segment in the (u, v) plane to find the area enclosed by a piece-wise loop (∂S):

$$Area(S) = \tag{36}$$
$$\frac{a^2}{8} \sum_i \left[\frac{\left(u_1^i - u_2^i \right) \left(\cos(2v_1^i) - \cos(2v_2^i) \right)}{\left(v_2^i - v_1^i \right)} + \frac{\left(v_1^i - v_2^i \right) \left(\cosh(2u_2^i) - \cosh(2u_1^i) \right)}{\left(u_1^i - u_2^i \right)} \right].$$

4.11. **Parabolic Coordinates.** A common mapping for the parabolic coordinate system is

$$\mathbf{G}\,(u, v) = \begin{pmatrix} uv \\ \frac{1}{2} \left(v^2 - u^2 \right) \end{pmatrix}$$

for $u, v \in \mathbb{R}$. The area 2-form is given by

$$d\omega = \left(u^2 + v^2 \right) \; du \wedge dv$$

where a suitable choice for ω is $\omega = \frac{1}{3} \left(u^3 \; dv - v^3 \; du \right)$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = \frac{1}{3} \int_0^1 \left[\left(v_2^i - v_1^i \right) \left(u_1^i + \left(u_2^i - u_1^i \right) t \right)^3 - \left(u_2^i - u_1^i \right) \left(v_1^i + \left(v_2^i - v_1^i \right) t \right)^3 \right] dt,$$

and sum over each linear segment in the (u,v) plane to find the area enclosed by a piece-wise loop (∂S):

(37)
$$Area(S) = \frac{1}{12}\sum_i \Big[\left(u_1^i + u_2^i\right)\left(v_2^i - v_1^i\right)\left(u_1^{i^2} + u_2^{i^2}\right) + \left(v_1^i + v_2^i\right)\left(u_1^i - u_2^i\right)\left(v_1^{i^2} + v_2^{i^2}\right)\Big].$$

4.12. **Hyperbolic Coordinates.** The hyperbolic coordinate system is commonly defined by the mapping

$$\mathbf{G}\left(u,v\right) = \begin{pmatrix} ve^u \\ ve^{-u} \end{pmatrix}$$

for $u \in \mathbb{R}$ and $v > 0$. The area 2-form is given by

$$d\omega = 2v\ du \wedge dv,$$

where a suitable choice for ω is $\omega = -v^2\ du$. Following §3.1, we integrate ω over $\boldsymbol{\gamma_i}$

$$\int_{\boldsymbol{\gamma_i}} \omega = -\int_0^1 \Big[\left(u_2^i - u_1^i\right)\left(v_1^i + \left(v_2^i - v_1^i\right)t\right)^2 \Big] dt,$$

and sum over each linear segment in the (u,v) plane to find the area enclosed by a piece-wise loop (∂S):

$$Area(S) = \frac{1}{3}\sum_i \Big[\left(u_1^i - u_2^i\right)\left(v_1^{i^2} + v_1^i v_2^i + v_2^{i^2}\right)\Big]. \tag{38}$$

4.13. **A 3-D case: Spherical Coordinates.** In this section we develop a formula for the *volume* enclosed by a surface, assuming spherical curvature. We follow the formalism and notation established in §3.2. The spherical coordinate system is defined by the mapping

$$\mathbf{G}\left(\rho,\theta,\phi\right) = \begin{pmatrix} \rho\sin\phi\cos\theta \\ \rho\sin\phi\sin\theta \\ \rho\cos\phi \end{pmatrix} \tag{39}$$

for $\rho \in [0,\infty)$, $\phi \in [0,\pi]$, and $\theta \in [0,2\pi)$. In this coordinate system, the volume 3-form is given by

$$d\omega = \rho^2 \sin\phi\ d\rho \wedge d\phi \wedge d\theta,$$

where a suitable choice for ω is $\omega = \rho^2\theta\sin\phi\ d\rho \wedge d\phi$. Suppose ∂D is covered by a triangular mesh, where the ith triangle T^i has vertices

$$\boldsymbol{a^i}, \boldsymbol{b^i}, \boldsymbol{c^i} \equiv \left(\rho_1^i, \phi_1^i, \theta_1^i\right), \left(\rho_2^i, \phi_2^i, \theta_2^i\right), \left(\rho_3^i, \phi_3^i, \theta_3^i\right). \tag{40}$$

Note, $\boldsymbol{a^i}, \boldsymbol{b^i}, \boldsymbol{c^i}$ must be cyclically ordered so as to induce a positive orientation on the surface (i.e. outwards normal vector). We parametrize the flat triangular region in (ρ,ϕ,θ) space as

$$\boldsymbol{\gamma_i}\left(t_1,t_2\right) = \boldsymbol{a^i} + \left(\boldsymbol{b^i} - \boldsymbol{a^i}\right)t_1 + \left(\boldsymbol{c^i} - \boldsymbol{a^i}\right)t_2$$

for $0 < t_1 < 1 - t_2$ and $t_2 \in \mathbb{R}$. Note, $\boldsymbol{G}\left(\boldsymbol{\gamma_i}\left(t_1,t_2\right)\right)$ maps the flat triangular region in (ρ,ϕ,θ) space to a spherical patch in (x,y,z) space. We integrate the 2-form ω

over ∂D, summing over all T^i thus forming a closed surface:

$$
\begin{aligned}
(41) \quad Vol(P) &= \sum_i \int_{T_i} \omega \\
&= \sum_i \int_0^1 \int_0^{1-t_2} \Bigg\{ \left(\rho_1^i + \Delta\rho_{21}^i t_1 + \Delta\rho_{31}^i t_2\right)^2 \left(\theta_1^i + \Delta\theta_{21}^i t_1 + \Delta\theta_{31}^i t_2\right) \\
&\qquad \sin\left(\phi_1^i + \Delta\phi_{21}^i t_1 + \Delta\phi_{31}^i t_2\right) \left(\Delta\rho_{21}^i \Delta\phi_{31}^i - \Delta\rho_{31}^i \Delta\phi_{21}^i\right) \Bigg\} \, dt_1 dt_2
\end{aligned}
$$

where $\Delta q_{jk} \equiv q_j - q_k$ for a generalized coordinate q. A closed form solution is found to exist using the mathematical analysis program, `Mathematica`. Integrating Eq. 41 numerically, however, is computationally straightfoward.

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